🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 · STANDARD 6.DS.6D

Appropriate Measures

Lesson 8-4 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 1 Support

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can choose the best measure of center for a data set based on its shape.

Language Objective: I can explain my choice using the words mean, median, outlier, and skewed.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Mean | Median | Outlier | Skewed | Data distribution | Variability

👆 *Tap any word to see what it means and a picture.*

1. The average of a data set is the ____________.
2. The middle value when data is in order, often best when there is an outlier, is the ____________.
3. A value far from the rest of the data, which can pull the mean, is an ____________.
4. Data that is bunched on one side with a tail on the other is ____________.
5. The way data values are spread out is the ____________.
6. How much the data values differ is the ____________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

As you sort scenarios into 'Use Mean' or 'Use Median,' what clue tells you a data set needs the median instead of the mean?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Mean** — The average. Add all the numbers, then divide by how many there are.
  Media — El promedio. Suma todos los números y divide entre cuántos hay.
- ☐ **Median** — The middle number when you put them in order.
  Mediana — El número del medio cuando los pones en orden.
- ☐ **Outlier** — A number that is much bigger or smaller than the rest.
  Valor atípico — Un número mucho mayor o menor que los demás.
- ☐ **Skewed** — When most data sits on one side with a tail on the other.
  Sesgado — Cuando la mayoría de los datos está de un lado con una cola del otro.
- ☐ **Data distribution** — How the data looks: where it sits and how spread out it is.
  Distribución de datos — Cómo se ven los datos: dónde están y qué tan separados están.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

I use the median when the data has ____.

Uso la mediana cuando los datos tienen ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: An outlier can pull the mean away from what is typical, making the median a better choice.

Evidence: Students choose the median (23) and identify 58 as an outlier that pulls the mean up to 27.6, making it unrepresentative.

Reasoning: This evidence connects the definition of mean to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **I use the median when the data has ____.**
  Uso la mediana cuando los datos tienen ____.
- **I use the mean when the data is ____.**
  Uso la media cuando los datos son ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

---

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Mean
2. **Notes 2:** Median
3. **Notes 3:** Outlier
4. **Notes 4:** Skewed
5. **Notes 5:** Data distribution
6. **Notes 6:** Variability
7. **Watch for this mistake:** *A common mistake in Appropriate Measures is defaulting to the mean just because "it uses all the data," without first checking whether an outlier or skewed shape is present. For example, given a player's points per game 14, 16, 15, 17, 16, 58, a student calculates the mean (136 ÷ 6 ≈ 22.7) and reports that as the typical game — but 58 is far from the cluster of 14–17, so it pulls the mean above 5 of the 6 games. The median (16) actually represents a typical game here. Before choosing a measure, always scan the data for an outlier or a long tail first: only use the mean once you've confirmed the data has neither.*
8. **Try It 1:** A. Mode, because it names the value that appears most — *The mode is the value that shows up most often. Here size M appears 4 times, so the mode (M) tells the manager what to order most.*
9. **Try It 2:** A. Range, because it is the highest minus the lowest value — *Range = max − min = 18 − 6 = 12. Because it uses only the two most extreme values, one outlier (18) changes the range a lot.*